Hybrid Anomaly Detection and LLM-Assisted Document Verification for Accounting Oversight of Notarial and Registry Offices

Hybrid Anomaly Detection and Document Verification

MARCELO LISBOA ROCHA*
Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, mlisboa@uft.edu.br

ELAINE TEIXEIRA PEDRO
Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, elaine.pedro@uft.edu.br

LUCIANO TENYLSON NOGUEIRA COSTA
Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, luciano.tenylson@uft.edu.br

RENATA DOS SANTOS RODRIGUES IDALGO
Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, renata.idalgo@uft.edu.br

DAVID NADLER PRATA
Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, ddnprata@uft.edu.br

GENTIL VELOSO BARBOSA
Postgraduate Program in Governance and Digital Transformation, Federal University of Tocantins (UFT), Brazil, gentil@uft.edu.br

The digitalization of administrative and financial processes in the public sector requires scalable computational mechanisms for oversight and exception prioritization. This study presents a hybrid artificial intelligence architecture for anomaly detection and document verification in the accounting records of Brazilian notarial and registry offices. The approach combines eight complementary anomaly detectors, consensus-based prioritization, optical character recognition, and document-level verification with a large language model. To preserve accounting heterogeneity, the quantitative models are executed within chart-of-accounts segments. In the experiment, 12,966 analytical occurrences from June to November 2025 were screened and 627 cases met the operational consensus criterion, corresponding to a retained review queue of 4.84%. A manual examination of 41 prioritized cases confirmed 34 inconsistencies in that examined subset. Complementary segmentation experiments showed narrower category-specific IQR limits and, in the clustering comparison, no outliers in the unsegmented run versus 27 outliers among 600 occurrences in the segmented run. The documentary module compares date, monetary value, and payer identification against OCR-extracted text, while authorized specialists retain responsibility for accounting and legal conclusions. The architecture therefore supports traceable, risk-oriented oversight without delegating administrative decisions to the language model.

corresponding author: Marcelo Lisboa Rocha, mlisboa@uft.edu.br. Postal address: Quadra 604 Sul, Al. 12, no. 29, Palmas/TO, Brazil, 77022-032. telephone: +55 63 98405-5547

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1 INTRODUCTION

Digital transformation in public administration has accelerated the transition from manual procedures to integrated computational systems designed to improve institutional efficiency and transparency [1]. In Brazilian notarial and registry offices, the inspection of administrative books must address large and heterogeneous volumes of financial information. Conventional sampling becomes increasingly limited when the number and diversity of records exceed the capacity of routine human review [2].

Outliers are observations that deviate substantially from prevailing data patterns. In accounting datasets, such deviations may indicate posting errors, operational failures, or transactions that warrant further investigation [3, 4]. Their value lies in directing scarce technical resources toward exceptional cases. Static rule systems, however, may generate excessive alerts and increase reviewer workload, which motivates the combination of complementary detectors and stronger evidence thresholds [5].

This study evaluates a hybrid architecture for intelligent oversight of extrajudicial offices. The pipeline combines consensus among multiple outlier-detection algorithms, segmentation by chart of accounts, optical character recognition (OCR), and semantic verification by a large language model (LLM). The central hypothesis is that accounting segmentation preserves category-specific distributions and improves the identification of relevant deviations when compared with a single global model [4]. The contribution is an operational, traceable workflow supported by comparative evidence from statistical IQR and unsupervised clustering experiments, in which quantitative screening and documentary comparison support, but do not replace, expert judgment.

2 THEORETICAL BACKGROUND

This section summarizes the concepts that support the proposed architecture: digital governance, statistical anomaly detection, and semantic interpretation of accounting documents.

2.1 Digital Governance and Intelligent Oversight

Digital governance treats information as institutional infrastructure and uses computational resources to improve the timeliness, transparency, and auditability of public-sector processes [1, 6]. Effective public governance also depends on transparency and accountability, so that decisions can be monitored and justified to society [7]. In this context, risk-oriented oversight can prioritize unusual transactions that might be missed by manual review alone [4, 8].

2.2 Outlier Detection in Accounting Data

An outlier or anomaly is an observation with a substantial deviation from the overall distribution or from dominant patterns in a dataset [4, 9]. In financial auditing, identifying such points helps isolate rare events that may be associated with errors or suspicious activity [4, 8]. More broadly, machine-learning methods have also been used to distinguish cloned social-media accounts from legitimate profiles, illustrating how learned behavioral and structural signals can support the identification of deceptive or atypical entities [18].

The interquartile range (IQR) is a robust statistical technique for flagging values outside the central dispersion of a distribution [10]. Because the thresholds are derived from quartiles, the method is less influenced by extreme magnitudes than mean- and standard-deviation-based rules [10, 11]. For heterogeneous accounting data, local or segmented analysis is especially relevant because the expected range of one expense category may be inappropriate for another [4].

2.3 OCR and Semantic Interpretation of Documents

OCR converts non-editable documents into machine-processable text and supports information extraction from governmental and legal records [12, 13]. Its practical use is affected by heterogeneous document layouts, noisy scans, spurious tags, and format variation [14, 15]. LLMs can add semantic comparison capabilities to the extracted text, helping reviewers trace structured accounting attributes back to documentary evidence [6, 16]. In document-image workflows, Meijering-filter-based processing has also been investigated for detecting document recapture, a condition that can affect OCR reliability and downstream verification [17]. Hybrid retrieval strategies can further improve document-grounded language-model workflows by selecting relevant evidence before semantic interpretation [19].

3 METHODOLOGY

The study is applied technological research using a hybrid quantitative and qualitative approach. The workflow comprises data preparation, segmented anomaly detection, consensus-based prioritization, OCR extraction, document-attribute verification, and specialist review.

3.1 Context and Data Corpus

The data originated from administrative books and accounting submissions of extrajudicial offices. Records were aggregated by office, chart-of-accounts category, and month/year. The experimental dataset contained 12,966 analytical records covering June through November 2025. CSV files and database records were integrated, and the TOTAL_MES variable was cleaned before modeling. Data preparation used Python 3.14.1 and Pandas 3.0.1.

The segmentation strategy preserves the internal statistical behavior of each accounting category. Each analytical unit represents a combination of office, account category, and month, while documentary evidence consists of invoices, receipts, and payment slips associated with prioritized cases.

3.2 Quantitative Module and Outlier Detection

The quantitative module analyzes TOTAL_MES with eight complementary detectors: IQR, Isolation Forest, Local Outlier Factor (LOF), One-Class Support Vector Machine (One-Class SVM), k-Nearest Neighbors (KNN), Minimum Covariance Determinant (MCD), Principal Component Analysis (PCA), and Cluster-Based Local Outlier Factor (CBLOF). The name CBLOF is used for the detector previously described as clustering. All

detectors receive the same 12,966 units and are executed separately within each chart-of-accounts segment.

Detector implementations were executed in Python 3.11 with PyCaret 3.3.2. For each of the eight detectors, Table 1 reports the implementation and parameter settings, the rule used to convert the detector output into a binary anomaly vote, the minimum segment size, and the treatment of ineligible segments or execution failures. Parameters not explicitly reported in Table 1 were retained at the standard values provided by PyCaret for the corresponding method. A vote of 1 denotes an anomaly and a vote of 0 denotes a non-anomaly under the detector-specific rule. When a detector cannot produce a valid output, its result is recorded as unavailable and is not counted as a negative vote, because a missing result does not establish that an occurrence is normal.

Table 1: Reported configuration of the quantitative detectors

Detector	Implementation and parameters	Score-to-label rule	Minimum segment size	Failure handling
IQR	LI = Q1 − 1.5 × IQR; LS = Q3 + 1.5 × IQR	Vote = 1 when TOTAL_MES < LI or TOTAL_MES > LS; otherwise vote = 0.	3470.50	If quartiles cannot be computed, record output as unavailable; do not assign vote = 0.
Isolation Forest	contamination = 0.05;	Vote = 1 when the model prediction is −1; otherwise vote = 0.	3470.50	If fitting or prediction fails, record output as unavailable.
LOF	n_neighbors = 20	Vote = 1 when the model prediction is −1; otherwise vote = 0.	At least n_neighbors + 1 observations. 11718	If the segment is smaller than required, record output as unavailable.
One-Class SVM	kernel = linear	Vote = 1 when the model prediction is −1; otherwise vote = 0.	4883.51	If fitting fails, record output as unavailable.
KNN	k = 5, or smaller when fewer than five records are available	Vote = 1 when the anomaly score is at or above 0.6; otherwise vote = 0.	3470.50	If no valid neighbor configuration is possible, record output as unavailable.
MCD	support_fraction = 0.5;	Vote = 1 when the robust-distance score is at or above 0.6; otherwise vote = 0.	3470.5	If covariance estimation fails or is singular, record output as unavailable.
PCA	n_components = None	Vote = 1 when reconstruction error is at or above 0.6; otherwise vote = 0.	5	If fitting or transformation fails, record output as unavailable.
CBLOF	At least 13 records;	Vote = 1 when the CBLOF score is at or above 0.6; otherwise vote = 0.	13 observations. 3989	If convergence fails, record output as unavailable; do not count it as a normal vote.

Source: Authors (2026).

3.3 Hybrid Architecture

The proposed architecture integrates quantitative and qualitative analysis. The quantitative level answers where an anomaly may exist by consolidating the eight detector outputs. For each eligible detector, its output is converted into a binary vote: 1 indicates anomaly and 0 indicates non-anomaly, according to the detector-specific rule reported in Table 1. The model_score is the sum of available anomaly votes. Unavailable outputs are excluded from the denominator and are not converted into negative votes. A record is prioritized when model_score is at least 3; the number of available detector outputs is retained for auditability. Records with at least three anomaly votes were prioritized for subsequent inspection. The three-vote threshold was adopted as an operational prioritization rule. A formal sensitivity analysis across alternative vote thresholds was outside the scope of the present study and should be addressed in future validation using the same manually reviewed

reference standard.

The qualitative level addresses the limits of numerical screening. OCR-derived text from attached documents enriches the prioritized records, and the LLM compares structured accounting attributes with information recovered from that text. The checks cover the document date, monetary value, and payer identification. The LLM does not autonomously determine accounting regularity, legal compatibility with the chart of accounts, or administrative outcomes. These conclusions remain the responsibility of specialists.

Together, the modules operate as a continuous pipeline: numerical analysis prioritizes suspected points, documentary analysis organizes supporting evidence, and human review determines whether a case is regular, irregular, or requires further diligence. Figure 1 presents the resulting human-in-the-loop architecture.

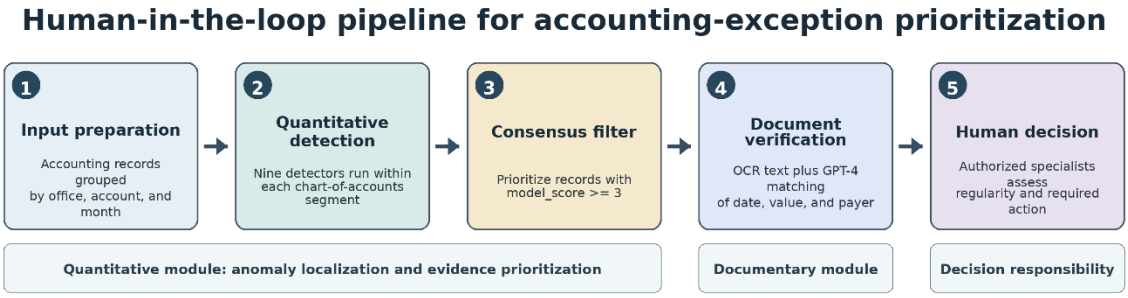


Figure 1: Human-in-the-loop pipeline for accounting-exception prioritization. Source: Authors (2026).

3.3.1 GPT-4 Configuration and Use for Document-Attribute Verification

Document verification used OpenAI GPT-4, registered in the institutional environment as the June 2024 version. Access occurred through tjro_llm, a package developed by the Court of Justice of the State of Rondonia that acts as an institutional proxy for models made available under an agreement with the Federal Data Processing Service (SERPRO).

Requests used a temperature of 0.1, top_p equal to 1, and a maximum limit of 8,192 tokens; all remaining parameters retained the environment defaults. For each document, OCR-extracted text and structured submission attributes were sent to the model. The response schema separately recorded whether the date, value, and payer identification matched and captured the supporting text for each check.

The schema also allowed missing fields, insufficient evidence, extraction failure, and non-interpretable responses to be recorded. Interpretable results were organized as 0 of 3, 1 of 3, 2 of 3, or 3 of 3 matching fields. The present study did not include a separately annotated reference sample for estimating OCR/LLM field-level accuracy. Therefore, the reported documentary outputs should be interpreted as automated correspondence or non-correspondence indicators, rather than validated estimates of accuracy, precision, recall, or F1-score. This coding reports automated correspondence between structured attributes and OCR text. Missing fields, insufficient evidence, OCR failures, and non-interpretable outputs were treated as absence of correspondence.

The complete prompt, system instructions, examples, structured response schema, and conversion rules should be preserved in supplementary material exactly as executed. GPT-4 outputs were used only to extract, organize, and present documentary evidence. The model did not determine legal or accounting regularity and did not produce autonomous administrative decisions.

3.4 Qualitative Module and Documentary Validation

Prioritized cases enter the OCR stage, in which payment slips, receipts, and invoices are converted into machine-readable text by the institutional PRISMA microservice using Tesseract. The resulting text is then submitted to the configured GPT-4 workflow described in Section 3.3.1. Objective field-level outputs create a traceable bridge between the numerical anomaly and the documentary evidence available to the human reviewer.

3.5 Evaluation Criteria

The operational evaluation examines the size of the initial analytical universe, the number of cases retained after consensus, the observed confirmation proportion and its 95% confidence interval in the manually reviewed prioritized subset, and the traceability of the resulting alerts. Non-prioritized records were not manually reviewed in the present study.

Table 2: Evaluation criteria and reported outcomes

Indicator	Result or description
Record source	Administrative books and accounting submissions, especially the Auxiliary Daily Book of Revenue and Expenses.
Initial analytical volume	12,966 records analyzed from June to November 2025.
Consensus criterion	Prioritization when model_score is at least 3.
Prioritized cases	627 cases retained for documentary examination and specialist review.
Retained review queue	4.84% of the initial universe; descriptive candidate-volume reduction of 95.16%.
Manual examination	Forty-one prioritized cases were manually reviewed by one specialist with substantial experience in the Tribunal's specific oversight area. The use of a single specialist reflected the limited availability of qualified personnel and the working-hour constraints associated with this specialized activity.
Interpretability	Detector votes, variables, and documentary evidence remain available for specialist review.

Table 2 summarizes the operational evaluation context rather than a test-set performance comparison. Of the 12,966 analytical records, the three-vote consensus criterion retained 627 cases (4.84%), producing a descriptive reduction of 95.16% in the documentary-review queue. The 41 manually examined prioritized cases provide descriptive evidence on the yield of focused review; however, because review was restricted to prioritized cases and was conducted by one specialist, these results should not be interpreted as estimates of accuracy, sensitivity, specificity, false-negative rate, or comparative performance for non-prioritized records. Accordingly, the table documents the scale, selection rule, and auditability of the prioritization process, while population-level validation remains future work.

3.6 Comparative Segmentation Experiments

The effect of chart-of-accounts segmentation was evaluated under two complementary analytical paradigms. The first experiment compared a classical IQR rule applied to the complete dataset with the same rule applied separately within each accounting category. The second used unsupervised clustering to compare runs with and without segmentation. Together, the experiments test whether the behavior attributed to segmentation is

limited to a statistical threshold or also appears in a computational model that captures structural similarity without prior labels.

The reported counts refer to distinct analytical scopes and are not used as a common denominator. The consensus pipeline screened 12,966 analytical occurrences. The IQR comparison used 29,942 records in the global scenario and 1,394 records in the Account 6 scenario to illustrate the effect of global and category-specific thresholds. The clustering comparison was a separate run involving 600 occurrences. Accordingly, the outlier counts from these analyses are descriptive and should not be interpreted as directly comparable performance measures.

3.7 Ethical Considerations and Anonymization

Identifiable attributes, including office-holder names and Brazilian taxpayer registration numbers, were removed or replaced with internal identifiers during preparation. The analytical workflow therefore focused on statistical and documentary patterns without unnecessarily exposing individuals. Access to unredacted evidence and decisions concerning regularity remained within the authorized oversight process.

4 RESULTS AND DISCUSSION

The hybrid architecture reduced the initial set of 12,966 analytical occurrences to 627 prioritized cases. This corresponds to a retained queue of 4.84% and a descriptive reduction of 95.16% in the number of candidate records forwarded to the next stage. The percentage describes volume reduction and should not be read as a directly measured reduction in time, cost, or human effort. For institutions handling large datasets, this filtering capacity supports more focused review than undirected sampling [2].

4.1 Model Performance and Validation

Manual examination of 41 prioritized cases, conducted by one specialist with substantial experience in the Tribunal's specific oversight area, confirmed 34 inconsistencies and did not confirm 7. This corresponds to an observed confirmation proportion of 82.93% (95% Wilson confidence interval: 68.74%–91.47%) in the examined subset. The use of a single specialist reflected personnel and working-hour constraints in this specialized activity. Therefore, the result is descriptive and does not provide independent inter-reviewer validation or a population-wide estimate for all prioritized records. The result shows that confirmed cases were concentrated in the evaluated priority set, but it does not by itself establish population-wide sensitivity, specificity, or false-negative rates.

The 627 prioritized cases are accounting occurrences selected by the quantitative consensus rule. The 2,341 documentary non-correspondences are field-level outputs produced during OCR/LLM comparison and may include more than one non-correspondence for a single prioritized occurrence, as well as unavailable or insufficient-evidence results. Under the field-level interpretation specified in Section 3.3.1, these outputs are treated as documentary non-correspondences or unavailable evidence rather than autonomous determinations of accounting irregularity. Specialist review remains necessary before an administrative conclusion is reached.

4.2 Global and Segmented Analytical Comparison

The comparative evaluation combines two perspectives: a classical statistical rule based on the interquartile range and an unsupervised clustering model. This dual design tests the effect of chart-of-accounts

segmentation under both distributional thresholds and similarity-based grouping. Table 3 consolidates the reported configurations and outcomes.

Table 3: Reported outcomes of the global and segmented experiments

Method	Scenario	Reported outcome
IQR	Global	29,942 records; IQR = 3,206.02; lower limit = -4,559.03; upper limit = 8,265.04; 3,439 outliers.
IQR	Account 6, Accounting Advisory	1,394 records; IQR = 1,098.65; lower limit = -1,089.97; upper limit = 3,304.62; 113 outliers.
Clustering	Without segmentation	No outliers were identified in the reported unsegmented run.
Clustering	With chart-of-accounts segmentation	27 outliers were identified among 600 occurrences.

4.2.1 IQR Segmentation Experiment

In the global scenario, the IQR method was applied to the complete dataset without accounting-category segmentation. The resulting interquartile range was 3,206.02, with a lower limit of -4,559.03 and an upper limit of 8,265.04. The method flagged 3,439 outliers among 29,942 records. The broad dispersion indicates that aggregating unlike accounting categories expands the statistical limits and can make the threshold permissive for category-specific deviations.

The category-level view in Figure 2 illustrates the effect of those global limits on Account 6, Accounting Advisory. Only the most extreme observations are highlighted when the category is evaluated under thresholds influenced by the broader heterogeneous distribution. When the IQR was calculated separately for Account 6, it decreased to 1,098.65, with a lower limit of -1,089.97 and an upper limit of 3,304.62. The segmented analysis flagged 113 outliers among 1,394 records. Figure 3 shows the resulting denser set of category-specific observations identified under the narrower limits. The comparison demonstrates that segmentation changes both the thresholds and the set of flagged records by preserving the internal distribution of the accounting category.

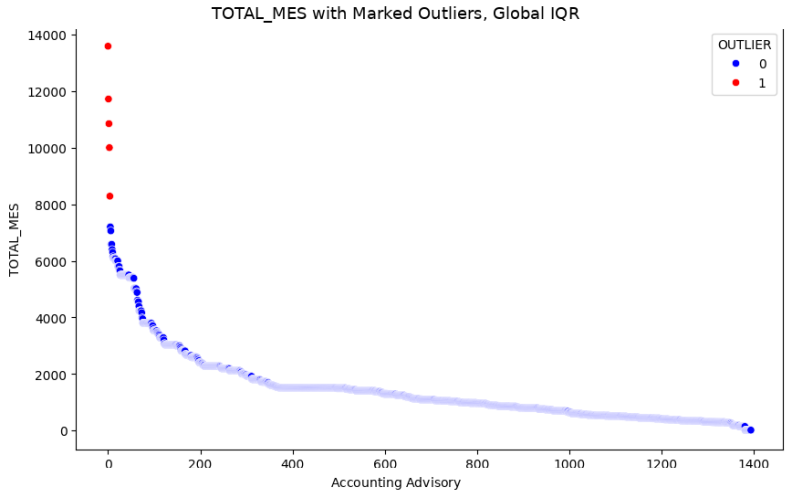


Figure 2: Accounting Advisory observations evaluated under the global IQR limits. Source: Authors (2026).

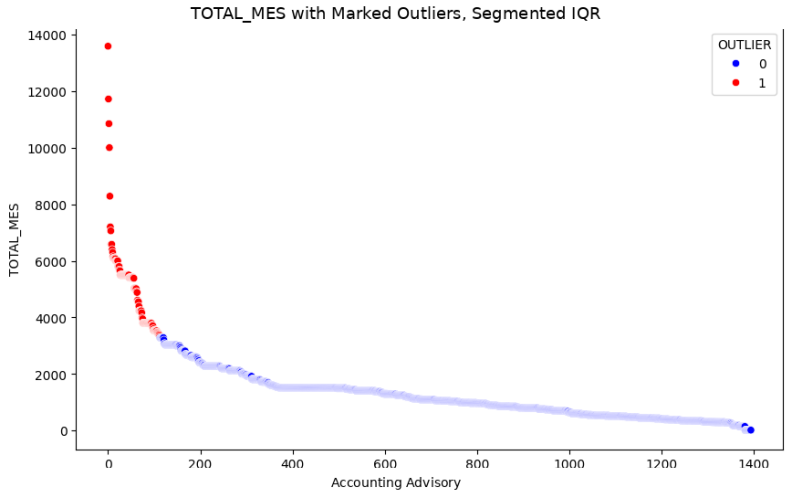


Figure 3: Accounting Advisory outliers identified with chart-of-accounts-segmented IQR limits. Source: Authors (2026).

4.2.2 Clustering Segmentation Experiment

The second experiment examined whether the segmentation effect also appeared in an unsupervised clustering model. In the run without chart-of-accounts segmentation, the model did not identify any outliers. After segmentation, it identified 27 outliers among 600 occurrences. The reported result indicates that the formation of more homogeneous category-specific groups increased the model's ability to discriminate anomalous patterns that were not signaled in the aggregated scenario.

Across both experiments, segmentation altered the analytical limits and the sets of occurrences selected for review. The comparison supports the interpretation that category-specific modeling preserves differences among expense types and changes detector sensitivity under both statistical and computational paradigms. Because the IQR scenarios cover different analytical scopes and the experiments do not provide complete ground-truth labels, the reported counts should not be interpreted as direct estimates of precision, sensitivity, or false-negative rates.

4.3 Case Analysis: Data-Protection Consulting

A representative case illustrates the value of combining quantitative and qualitative evidence. In the Accounting Advisory category, one occurrence received votes from all eight detectors. Human investigation found recurring payments recorded as data-protection consulting, totaling 53,533.90 Brazilian reais (BRL) in 2025, and concluded that the expense nature was incompatible with the accounting category used. The case shows how statistical consensus can direct specialists to materially relevant records while the final contextual judgment remains human [7].

5 CONCLUSION

This study presented a hybrid anomaly-detection and LLM-assisted document-verification architecture for accounting oversight of notarial and registry offices. The workflow combines eight quantitative detectors, chart-of-accounts segmentation, OCR, field-level document verification with GPT-4, and specialist review.

Segmentation avoids the statistical distortion that can arise when unlike expense categories are modeled together, while multimodel consensus provides an explicit rule for prioritizing records.

The experiment retained 627 of 12,966 analytical occurrences for the documentary stage and subsequent review. In the manually examined subset, 34 of 41 cases were confirmed as inconsistent. These results support the use of the architecture as a traceable prioritization mechanism, while the field-matching limits of the language model and the scope of the manual examination and the use of one specialist reviewer require cautious interpretation.

The comparative experiments further showed that segmentation materially changes detector behavior. For Account 6, the category-specific IQR decreased from the broad global value of 3,206.02 to 1,098.65, with 113 of 1,394 records flagged under the segmented limits. In the clustering comparison, the unsegmented run identified no outliers, whereas the segmented run identified 27 among 600 occurrences. These findings support category-aware modeling across distinct analytical paradigms, without substituting reported alert counts for fully labeled performance estimates.

Future work should expand empirical validation through independent assessment by multiple specialists, which would allow inter-rater agreement to be measured and would strengthen the generalizability of the manual-review findings. A fully labeled evaluation design that includes prioritized and non-prioritized records would also allow sensitivity, specificity, and false-negative rates to be estimated.

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